

3**B** $V_{\text{relative to ground}}$

$$= \sqrt{350^2 - 100^2} = 335 \text{ km/h}$$

$$V_{\text{plane}} = 350 \text{ km/h}$$

$$V_{\text{plane}} = 100 \text{ km/h}$$

 $V_{\text{relative to ground}}$ 

A vector diagram illustrating the relationship between the plane's velocity and its velocity relative to the ground. It features a right-angled triangle with blue arrows. A horizontal arrow at the top is labeled $V_{\text{plane}} = 100 \text{ km/h}$. A vertical arrow on the right is labeled $V_{\text{relative to ground}}$. A hypotenuse arrow on the left, pointing from the bottom vertex to the top-left vertex, is labeled $V_{\text{plane}} = 350 \text{ km/h}$.